

GEOG 172

Welcome to Intermediate Geographical Analysis

This course is designed to help students learn quantitative methods for the analysis of geographical data. Tutorial includes the use of statistical software for exploratory spatial data analysis.

Lecture 0

PATTERN & PROCESS

If you want to describe, predict, explain, or successfully intervene in geographic phenomena, you begin by examining spatial patterns because they provide clues to the processes that may be causing the phenomena.

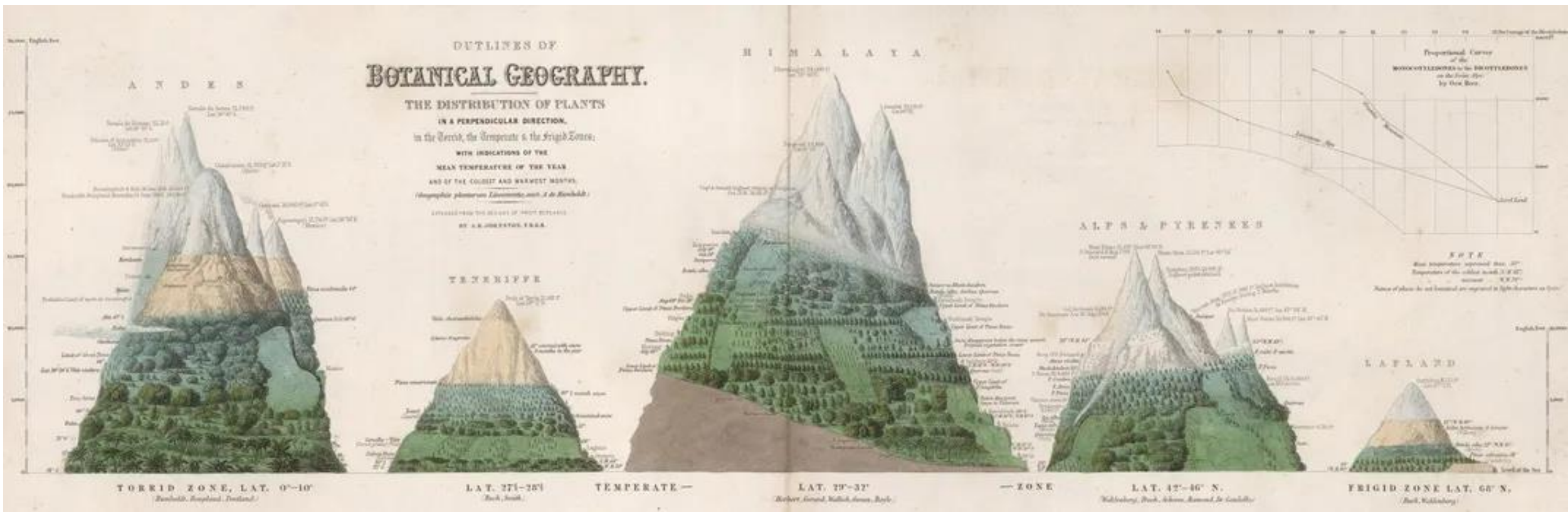


Figure 0.1 The Geographic Distribution of Plants in a Perpendicular Dimension

Detail from an 1850 atlas showing the distribution of plant life on several mountain ranges at different latitudes. Source: Johnston & von Humboldt (1850)

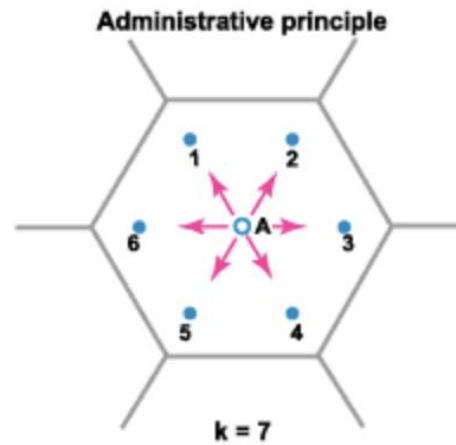
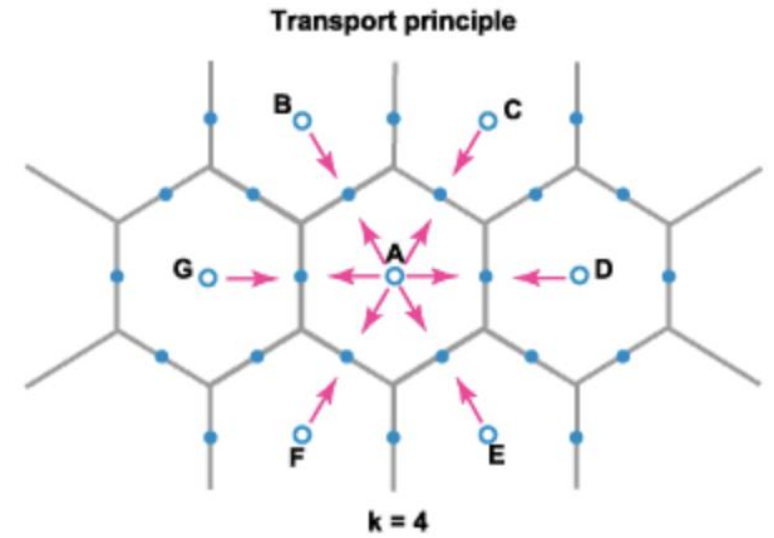
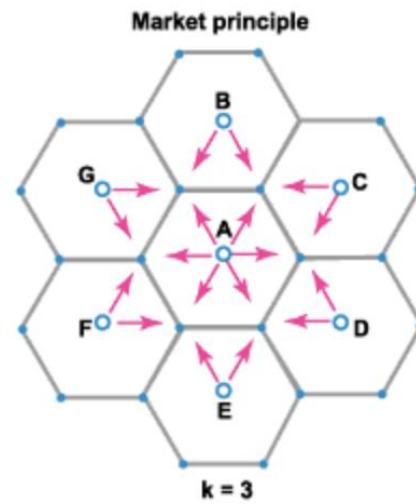
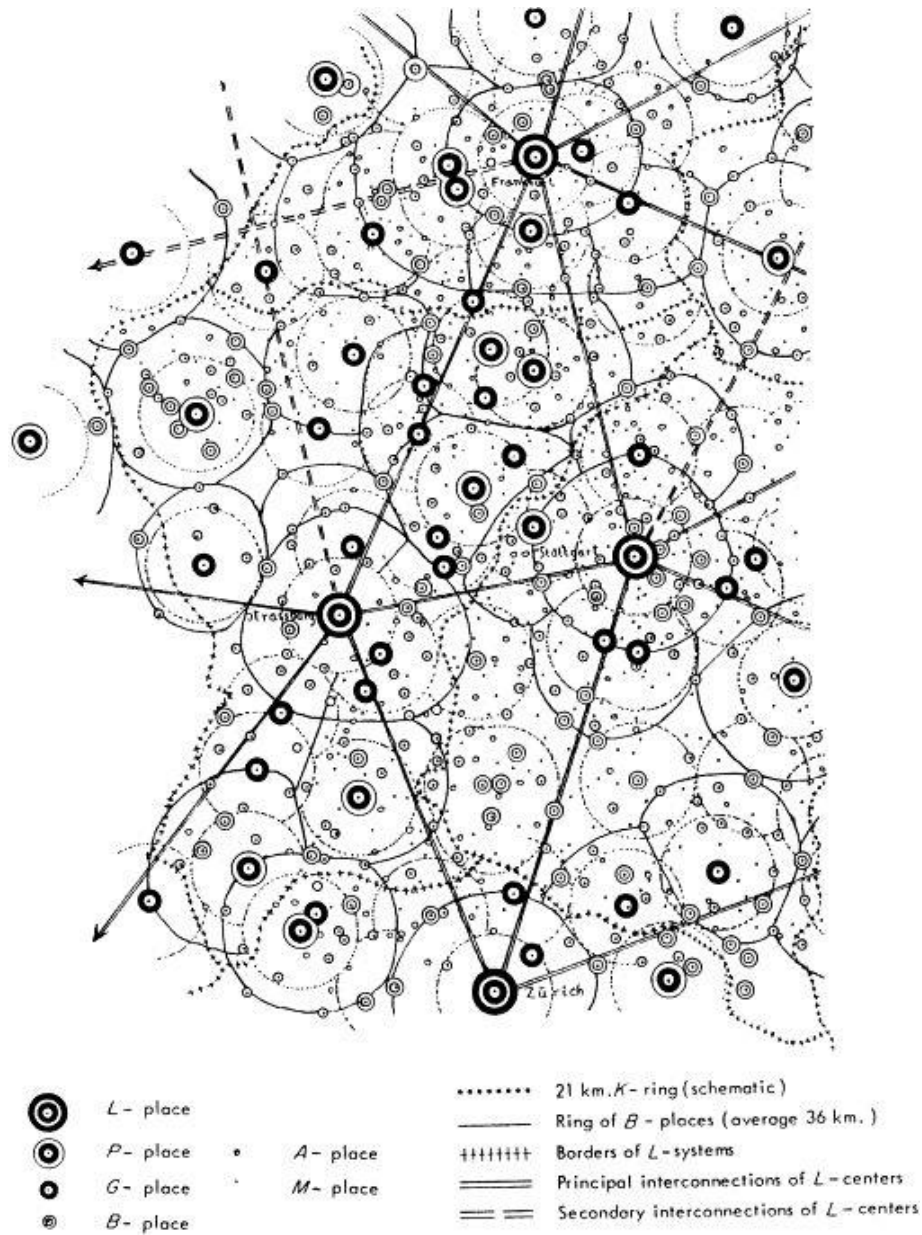


Figure 0.2 Christaller's Hierarchy of Settlements

GOULD'S (1970) CRITIQUE

Many of the procedures and assumption of classic statistics breakdown when analyzing spatial data

- *Geographic are not samples*
- *Geographic observations are almost never independent or random*
- *Because n is always large, you can almost always find statistically significant results*
- *What matters is scientific significance, not statistical significance*

Course Structure

To the internet ...

Your Instructors

... in absentia Die Hu

ACTIVITY

Overview: In this introductory activity, students collectively create a map of the geographic experience and curiosity represented in the classroom. Each student contributes two locations: a place in the United States they know well or especially enjoyed visiting, and a place they have never visited but would most like to go.

Purpose: Generate, identify, and describe simple spatial patterns

Task:

- Place a sticker on the class map identifying a place you know grew up, know particularly well, have lived, or especially enjoyed.
- Place a sticker a place in the United States that they have never visited but would most like to go

UPCOMING IN WEEK 1

On Tuesday, you are going to get acquainted with the nature of spatial data by participating in a pair of in-class exercises.

There will be **No Class Thursday** and you are asked to independently review some statistics you should already be familiar with (e.g., means, popular probability distributions, hypothesis testing).

There will be tutorial next week. The first tutorial is designed to familiarize you with the tutorial structure, R & R Studio, and the computational notebooks we will use during the course. We've prepared this as a platform for you to succeed.